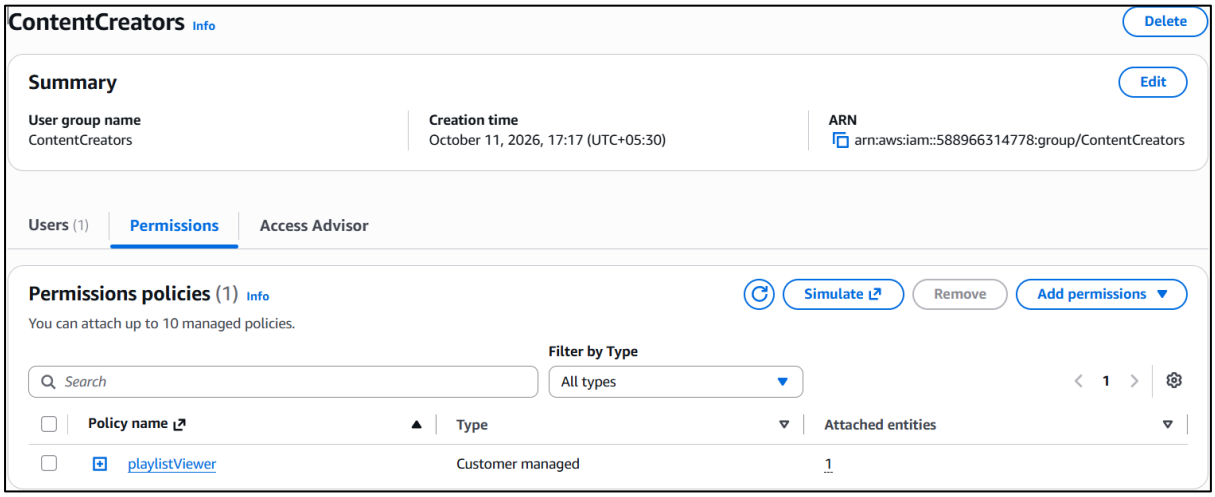
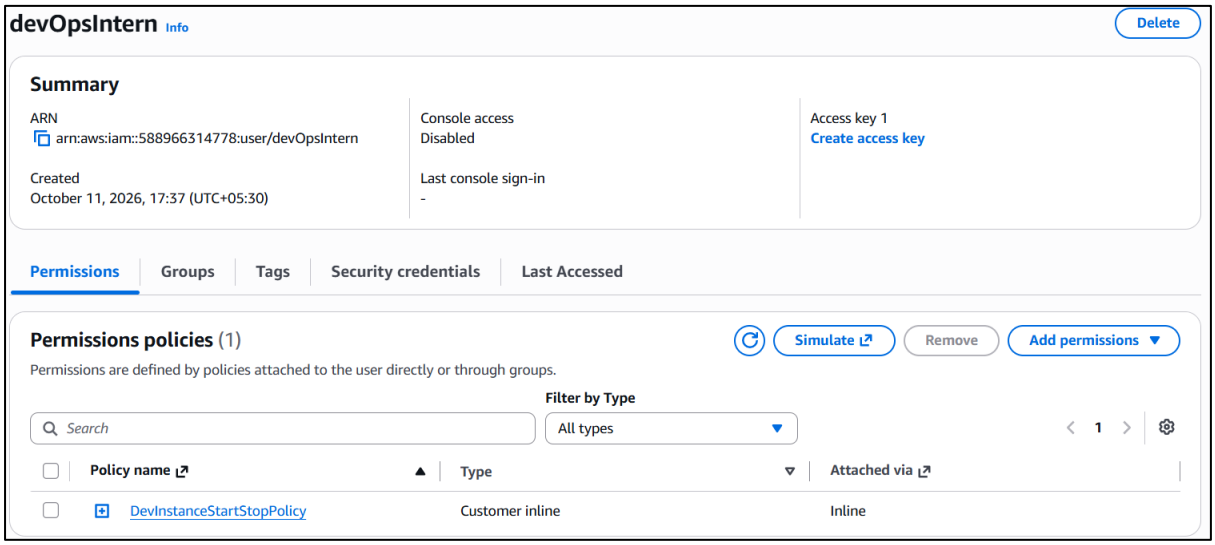
Security In The Cloud – Assignment-2

1. Create a custom JSON-based policy in AWS IAM that allows read-only access to all objects in an S3 bucket named 'my-music-playlists', and attach this policy to a new user called 'playlistViewer'.

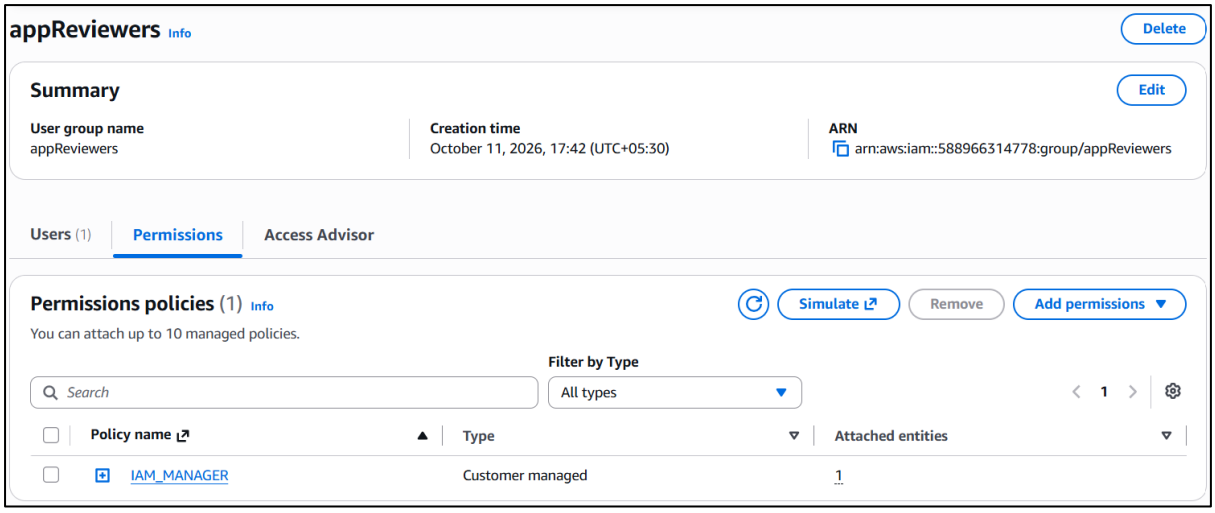


2. Write an inline policy in JSON that allows a user to start and stop EC2 instances only with the tag Environment: Dev, and attach it directly to an existing IAM user named 'devOpsIntern'. Hint: Use 'Condition' in your policy to filter by tag.



3. Create a managed policy in AWS IAM that grants permission to list and view DynamoDB tables, then attach this policy to a group called 'appReviewers' which could represent users reviewing app data like Zomato order history.

Security In The Cloud – Assignment-2



4. Given the following JSON policy, identify and fix the error that would prevent it from granting access as intended: { "Version": "2012-10-17", "Statement": [{ "Effect": "Allow", "Action": "s3:GetObject", "Resource": "arn:aws:s3:::user-uploads/*" }] } Hint: Check if the policy allows listing the bucket as well as getting objects.

Corrected JSON policy

```
{
 "Version": "2012-10-17",
 "Statement": [
   {
    "Effect": "Allow",
    "Action": "s3:ListBucket",
    "Resource": "arn:aws:s3:::user-uploads"
   },
   {
    "Effect": "Allow",
    "Action": "s3:GetObject",
    "Resource": "arn:aws:s3:::user-uploads/*"
   }
 ]
}
```

Explanation

- s3:ListBucket allows users to view the list of objects in the bucket. Its resource ARN refers to the bucket itself.
- s3:GetObject allows users to read or download objects. Its resource ARN refers to the objects inside the bucket.
- The original policy was missing the s3:ListBucket permission.

Security In The Cloud – Assignment-2

5. Use ChatGPT to generate a JSON IAM policy that allows users to only update their own profile information in a DynamoDB table named 'UserProfiles', then review and test the policy in the AWS Policy Simulator. Hint: Ask ChatGPT for a policy with a condition that enforces users can update only their own data.

